HexOct: Knitting Crystals

The edge-on slice of every close-packed stacking as a knitted pattern

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DICTO proposed the idea. The method and the generator are ours, made with DICTO; computation and proofs with Claude (Anthropic).
Version 2 cites the HexOct book by its concept DOI; the text is otherwise unchanged.

Abstract. DICTO proposed that a knitted fabric can be an exact edge-on slice through a stacking of balls. We prove that one plane does this for every close-packed stacking. The plane perpendicular to a row of touching balls (a {11-20} plane) meets every layer in one ball centre per repeat, at a third of the repeat apart for the three layer positions A, B, C, and cuts the pieces between the balls into just two shapes: each tetrahedron into an isosceles triangle and each octahedron into a rhombus. The mid-planes between the layers cut these into the halves of DICTO's Rule of Halves, and in every layer the tetrahedral and octahedral halves take exactly equal areas, so the share of blue (hexagonal) colour in the slice is the hexagonality itself. Knitting places each layer by bias rows: the row moves a third of a repeat across every gap, the same way again after a cubic layer and back after a hexagonal one, so cubic stretches knit as a straight slant and hexagonal ones as a chevron. A generator turns any stacking into a four-colour chart with row-by-row instructions; five swatches (4H, 3C, 2H, 6H and the Fibonacci stacking) are given in full.

Stack oranges in layers and cut the pile straight down with a very sharp knife. On the cut face you see the oranges in rows, each layer shifted a little to the side of the one below, and between them small triangles and diamonds of empty space. Colour the triangles and diamonds, and the cut face becomes a pattern a knitter can follow, stitch by stitch, row by row.

1. Context

Layers of balls. Equal balls packed as tightly as possible in a flat layer touch six neighbours each; the centres form a triangular grid. The next layer rests in the hollows of the first, and there are two kinds of hollow. With three positions in all, called A, B and C, every close-packed crystal is a word in these letters with no letter repeated next to itself: ABCABC… is the cubic (face-centred) packing 3C, ABAB… the hexagonal packing 2H, ABCB… the four-layer 4H, ABCACB… the six-layer 6H. A layer is *c* (cubic) when its two neighbours differ, as B in ABC, and *h* (hexagonal) when they agree, as B in ABA; the stacking is equally well written as its word of c and h. The share of h layers is its *hexagonality*: 0 for 3C, 1 for 2H, 1/2 for 4H, 1/3 for 6H.

The pieces. Between two layers the space left by the balls splits into tetrahedra (three balls below and one above, or one below and three above) and octahedra (three below and three above). DICTO's Rule of Halves (#57) cuts every such piece at the mid-plane between its two layers and gives the half nearer a layer to that layer; his HexOct tiles (#75) collect a layer's halves into one c-tile or h-tile. In the HexOct drawings the half of a c layer is gold and the half

of an h layer blue; a piece whose two layers differ in kind (a Janus piece) is half gold and half blue.

The Hexagonality Law. DICTO's law, in the form he approved, is a law of structural crystallography: in every close-packed crystal structure, however its layers are stacked, every count of the structure (its pieces, holes, bonds and swaps) is a straight line in the structure's hexagonality (#69 and the book *HexOct: The Hexagonality Law*).

DICTO's proposal (31 October 2026). Read the HexOct structure as a knitted fabric. The fabric is an exact edge-on slice through the stacking: stitch = ball, rows = layer; A, B and C appear as sideways offsets of 0, 1/3 and 2/3 of the repeat; the knitting places every layer by bias shifting (a decrease at one end of a row and an increase at the other move the row one stitch sideways); a c layer shifts the same way as the step before it and an h layer shifts back, so cubic knits as a straight bias slant and hexagonal as a chevron. Every area of the slice is coloured by the piece it belongs to, each half in its own layer's colour: gold for c, blue for h; tetrahedral pieces dark, octahedral pieces light.

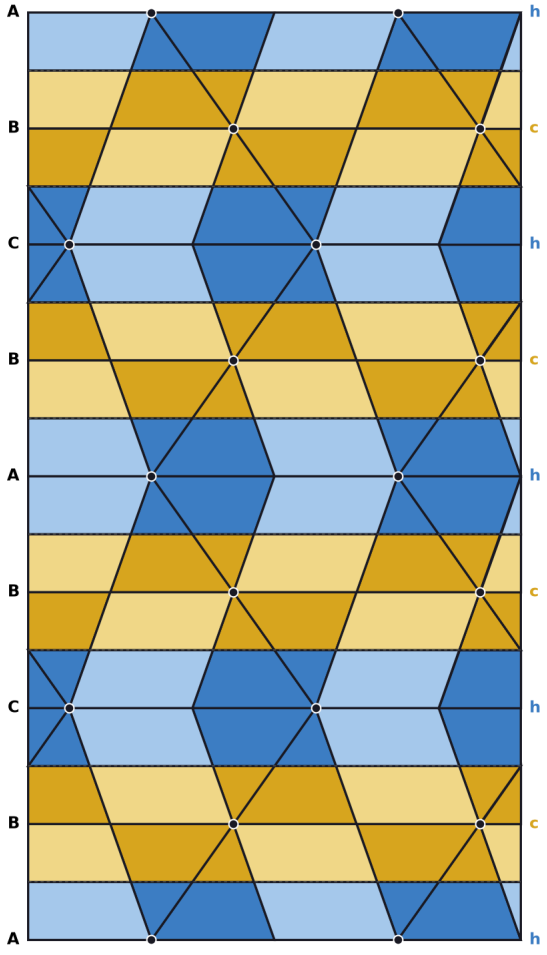


Figure 1. The exact slice of the 4H stacking ABCB, two periods, two repeats wide. Dots are ball centres; dashed lines are the mid-planes. Gold: halves of c layers; blue: halves of h layers; dark: tetrahedral; light: octahedral.

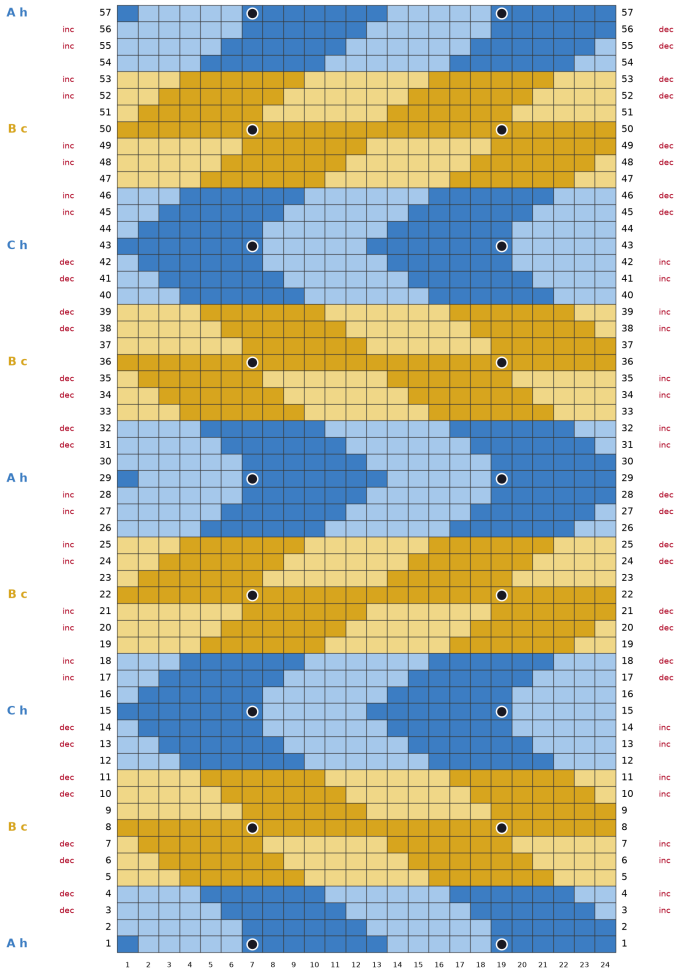


Figure 2. The same slice as a knitting chart, 24 stitches by 57 rows, in the knitted (fabric) grid: dots are the ball stitches, *dec*/*inc* mark the bias rows. Knitted, every bias row moves its row one stitch and the chart takes the shape of Figure 1.

2. Background: threads, knots, nets and folds

Before there were computers, a row of holes in a card told a loom which threads to lift; before there was knot theory, a theory of atoms asked which knots can exist. Threads have carried patterns, records and mathematics for thousands of years.

Knitting. Knitting makes a fabric from one continuous thread pulled through its own loops, row by row. The oldest surviving true knitting comes from Egypt, socks and fragments of about the 11th century AD [B1]. Older Egyptian socks, such as the red wool socks of the 4th–5th century AD from Oxyrhynchus in the Victoria and Albert Museum, look knitted but are *nålbinding*, an older technique that loops short lengths of yarn with an eyed needle [B2]; a 3rd-century fragment from Dura-Europos, once called the first knitting, is nålbinding too [B1]. Whether knitting grew out of nålbinding is not settled [B1].

Machines and programs. Tradition credits William Lee of Calverton, near Nottingham, with the stocking frame of 1589, the first knitting machine; the year and the details of his life rest on later accounts, not on records of the time [B3]. Joseph-Marie Jacquard's loom (patented 1804) chose the warp threads to lift from a chain of punched cards; Charles Babbage took the cards for his Analytical Engine, and Ada Lovelace wrote in 1843 that the Engine "weaves algebraical patterns just as the Jacquard-loom weaves flowers and leaves" [B4]. In 1995 Shima Seiki of Wakayama launched the first flat knitting machine that knits a whole garment in three dimensions without seams (WHOLEGARMENT) [B5].

Into space. The reflectors of large deployable antennas are a fine tricot-knit mesh of gold-plated molybdenum wire; this mesh flew on the deployable single-access antennas of NASA's Tracking and Data Relay Satellites [B6], and mesh reflectors of this kind (Northrop Grumman's AstroMesh, Harris) serve many communication satellites [B7]. The programs of the Apollo Guidance Computer were stored in core rope memory, made at Raytheon by women who threaded wires by hand through or around small magnetic rings: the work is called weaving, and it was not knitting [B8].

Nets. Netting ties a thread to itself in knots to make a mesh. The Antrea net from Karelia, of willow bast and about 8300 BC, is the oldest known fishing net [B9]. A net is a grid of knots, as a knitted fabric is a grid of loops.

Weaving, rugs and tapestries. Weaving interlaces two systems of threads, warp and weft, at right angles; knitting loops one thread. Grünbaum and Shephard began the geometry of woven fabrics with the *isonemal* fabrics, whose symmetries carry any strand to any other (1980), and Roth later classified their symmetry groups, the woven cousins of the crystallographic layer groups [B10]. A knotted-pile carpet ties a knot round the warp for every point of its design: the Pazyryk carpet in the Hermitage, from a burial in the Altai of the 5th–4th century BC and the oldest known pile carpet, has about 36 symmetrical knots per square centimetre [B11], and Persian and Anatolian rugs repeat their field patterns over grids of knots. Tapestry weaves the picture into the weft, as at the Gobelins manufactory in Paris, royal from 1662 [B12]; the Bayeux Tapestry, despite its name, is embroidery, wool stitched on linen [B13]. Carpets and tapestries are grids of knots or crossings that carry a picture; the fabric of this paper is a grid of stitches that carries an exact slice of a three-dimensional crystal.

The mathematics of knitted fabric. Grishanov, Meshkov and Omelchenko described knitted and woven fabrics as knots and links on a torus and built invariants for them [B14]; Morton and Grishanov added Alexander-type invariants [B15]. The elasticity of stockinette was explained from yarn bending and stitch topology by Poincloux, Adda-Bedia and Lechenault [B16], and Singal, Dimitriyev and co-workers showed that mixing knit and purl stitches programs how far a fabric stretches [B17]. belcastro and Yackel's books and Taimiņa's hyperbolic crochet are in Section 7.

Folding, the sister art. Origami turns a flat sheet into a solid by folds. Kōryō Miura devised the Miura fold in 1970 at the University of Tokyo's space institute (a related pattern appears in Yoshimura's 1951 study of buckling), and it flew in 1995 in the deployable solar-array experiment of Japan's Space Flyer Unit [B18]. Brigham Young University and JPL, with the origami artist Robert Lang, designed a solar array that folds to 2.7 m across and opens to 25 m [B19]; JPL's Starshade, a petal-shaped screen to block starlight, is stowed by an origami-based furling pattern [B20]. Knitting and folding both turn a line or a sheet into a solid form; here a knitted fabric carries an exact slice of a crystal.

Legend, then history. *Legend:* at Gordium in 333 BC Alexander faced the knot that tied the yoke of an old wagon; most writers, as Plutarch reports, say he cut it with his sword, but Aristobulus, quoted by both Arrian (Anabasis 2.3) and Plutarch (Alexander 18), says he pulled out the pin of the pole and freed the yoke [B21]. Ariadne's thread led Theseus out of the labyrinth. The three Moirai are named the spinner (Clotho), the allotter (Lachesis) and the unturnable (Atropos), and later tradition has them spin, measure and cut the thread of each life; one Norse poem (the first lay of Helgi Hundingsbani) shows the Norns twisting the threads of fate [B22]. In the Odyssey Penelope weaves a shroud by day and unpicks it by night to hold off her suitors (Books 2, 19 and 24). *History:* the Inca khipu records numbers in knots on hanging cords, in base ten, a reading established by Locke (1912); whether khipus also record narrative, calendars or astronomy is debated, with calendar readings proposed for some (Zuidema, Urton, Laurencich-Minelli and Magli) [B23]. The claim that Egyptian "rope-stretchers" laid out right angles with a 3-4-5 knotted rope goes back to Moritz Cantor (1882) and has no surviving evidence [B24]. The sailor's knot is a speed: a chip log, a line knotted every 47 feet 3 inches, was let run behind the ship for a 28-second sand-glass and its knots counted [B25]. Marshall Islands stick charts map ocean swells in sticks, with shells for islands, and were learnt by heart rather than carried at sea [B26]. In 1867, after Tait's experiments with smoke rings, William Thomson (Lord Kelvin) proposed that atoms are knotted vortices in the ether; to know which atoms could exist, Tait set out to list the knots by their crossings and published the first knot table in 1876. The vortex atom was abandoned; knot theory stayed [B27]. Knot theory was born from a theory of atoms, a fitting ancestor for DICTO's *Quanta of Space*, the pieces of space, and for a knitted crystal.

3. Main theorems

Ball spacing a = 1, layer spacing h = √(2/3). Take coordinates with the layers horizontal, the x-axis along a row of touching balls and the y-axis across it in the layer; then the A centres are (m + n/2, n√3/2, z), B = A + (1/2, √3/6, 0) and C = A + (1, √3/3, 0). Let P be the vertical plane x = 0, perpendicular to a row of touching balls: a {11-20} plane, seen along ⟨11-20⟩. On P use the sideways coordinate s = −y.

> **Theorem 1 (the slice).** For every close-packed stacking: (a) P meets each layer in exactly one ball centre per repeat √3, at s = 0, √3/3, 2√3/3 (0, 1/3, 2/3 of the repeat) for A, B, C;

every other ball centre lies at distance 1/2, so those balls touch P at one point only. (b) P is a mirror plane of the stacking. Every tetrahedron that P cuts meets it in an isosceles triangle with sides 1, √3/2, √3/2 (angles 70.53°, 54.74°, 54.74°) whose side 1 is a whole edge joining touching balls of the two layers; every octahedron that P cuts meets it in a rhombus of side √3/2 and angles arccos(1/3) = 70.53° and 109.47°, with two sides in the layer planes. (c) The mid-planes cut each triangle into a small triangle (1/2, √3/4, √3/4) at its tip and a trapezoid (√3/2, √3/4, 1/2, √3/4) at its base, and each rhombus into two congruent parallelograms (√3/2, √3/4): these are the halves of the Rule of Halves, and they tile the slice. (d) In the band of every layer (from the mid-plane below to the mid-plane above) the tetrahedral and the octahedral halves have equal area. So exactly half of the slice is dark and half light, and the blue share is exactly the hexagonality.

Proof. (a) A centre lies on P when m + n/2 = 0 for A, m + n/2 + 1/2 = 0 for B and m + n/2 + 1 = 0 for C, that is n even for A and C and n odd for B. Then y = n√3/2 runs through multiples of √3 for A, and y = n√3/2 + √3/6 (n odd) and y = n√3/2 + √3/3 (n even) give s ≡ √3/3 and 2√3/3 modulo √3 for B and C. All other centres have x a nonzero multiple of 1/2, so they are at least 1/2, one ball radius, from P. (b) x ↦ −x maps each layer's set of centres to itself (for A, (m, n) ↦ (−m−n, n); for B and C the same with the shift absorbed), so it is a symmetry of every stacking and of its tiling by pieces. A piece that P cuts in a region of positive area meets its own mirror image in that region, so it is its own mirror image, and P is one of its mirror planes. The mirror planes of a regular tetrahedron of edge 1 each contain one edge and the midpoint of the opposite edge, giving the triangle 1, √3/2, √3/2. A regular octahedron has mirror planes of two kinds: through four corners, cutting a square, and through two opposite corners and the midpoints of two opposite edges, cutting a rhombus of side √3/2 with diagonals √2 and 1. The octahedra of a close packing have a three-fold axis along the stacking direction, and the square planes, each perpendicular to a four-fold axis, are tilted to it; P is vertical, so it is of the second kind. (c) The mid-plane lies at half the height of every piece; in the triangle it cuts the two sides √3/2 at their midpoints, leaving a similar triangle of half size at the tip; in the rhombus it joins the midpoints of the two slanting sides. The pieces tile space, so their sections tile P, and so do their halves. (d) The triangle has area √2/4, its tip triangle √2/16 and its base trapezoid 3√2/16; the rhombus has area √2/2 and each parallelogram √2/4. Per repeat, a gap holds one area √3·h = √2; by the exact computation (Section 4) it holds one rhombus and two triangles, one standing on the lower layer and one hanging from the upper. The lower half of the gap is then a base trapezoid, a tip triangle and a parallelogram: 3√2/16 + √2/16 = √2/4 dark against √2/4 light, and the same for the upper half. So every half-gap, and every layer band, is half dark and half light; and every band has the same area √2 per repeat, so the blue share is the share of h layers. ∎

Theorem 2 (bias placement). Let a chart repeat be 3k stitches and move the knitting k stitches sideways across every gap, to the right when the stacking steps A→B, B→C or C→A and to the left otherwise. Then (a) the ball stitches of every layer fall in the same chart columns; and (b) the step out of a c layer goes the same way as the step into it, and the step out of an h layer goes back: a run of c layers knits as one straight slant, a run of h layers as a zigzag (chevron).

Proof. (a) By Theorem 1(a) a step to the next letter in the cycle A→B→C→A moves the ball centres by +1/3 of the repeat and a step back by −1/3; the chart moves by ±k stitches = ±1/3 of 3k, the same amount, so the true position of a ball stitch minus the accumulated shift is the same for all layers. (b) If layer k is h, layer k+1 repeats layer k−1, so the step out undoes the

step in. If it is c, layers k−1, k, k+1 are the three different letters, so the step out continues around the cycle in the same direction. ∎

So a stacking's word of c and h is read directly as its knitting: c means "keep going", h means "turn back". Over a cubic stretch the shifts add up, and a 3C fabric leans by k stitches per layer; a hexagonal stretch returns to its column every two layers.

4. Method and exact checks

Pieces. The pieces are the classical tetrahedra and octahedra of the close packing (the HexOct pieces at weight $\lambda = -1/8$ of #75, the ones DICTO's ring builds use). For each gap, up-tetrahedra, down-tetrahedra and octahedra are built from the exact centres; the section of each with P is computed in exact rational coordinates (s in units of $\sqrt{3}/6$, height in units of h), cut at the mid-plane, and coloured by its layer. Checks, all exact: the halves cover the window exactly once (4H: area 96 = 96 in these units); the whole sections are exactly the triangle and the rhombus of Theorem 1 (table of all sections with corners, sides and angles in `pieces-in-slice.json`); every layer band has equal tetrahedral and octahedral area for 4H, 3C, 2H, 6H and the Fibonacci stacking (`areas.py`).

Gauge. The true ratio of repeat to layer height is $\sqrt{3} : \sqrt{(2/3)} = 3/\sqrt{2} = 2.121$. With a repeat of 12 stitches (k = 4), 7 rows per layer and stitches 5 wide to 4 tall (a common stockinette gauge, e.g. 22 stitches and 28 rows to 10 cm), the chart gives $12 \times 1.25 / 7 = 2.143$, 1.0 % wide. The ball row of layer k is row 7k + 1, so each mid-plane falls on a row boundary and the halves change between rows. Across each gap the 4 bias rows are gap rows 2, 3, 5 and 6, symmetric about the mid-plane.

Chart. Each stitch takes the colour that covers most of it (4 × 4 sample points); where two colours cover exactly half each, which happens mainly in the ball rows, the dark shade is taken. This choice changes shade only, never hue, and it is why the charts count somewhat more dark than light stitches although the exact areas are equal (Theorem 1(d)). The chart is drawn in the knitted grid: the true position of a stitch is its chart column plus the shift accumulated by the bias rows. The 24-stitch window starts half a repeat after a ball, so the ball stitches, knitted as bobbles, sit at stitches 7 and 19, away from the edges. A knitter needs four colours: dark gold, light gold, dark blue, light blue (a purely cubic or purely hexagonal stacking needs two).

5. The generator

`hexknit.py` takes a stacking as a word of c and h or of A, B and C (repeated) and a number of layers, and writes the exact slice, the chart, the row-by-row instructions (colour runs in working order, bias edges, bobbles) and a printable swatch sheet:

```
hexknit.py --ch hchc --layers 5 --name 4H · --abc ABC --layers 7 --name 3C · --ch chcchchcchcch --layers 13 --name Fibonacci
```

Its 4H output is identical, row by row, to the first 4H swatch sheet, made before the generator.

6. Five swatches

Swatch	Stacking (c/h)	Layers	Stitches × rows	Colours	Character
4H	ABCBA (hchch)	5	24 × 29	4	alternating gold and blue bands; R R then L L: a zigzag of period four layers
3C	ABCABCA (all c)	7	24 × 43	2 (gold)	bias always right: one straight slant, 24 stitches over the swatch
2H	ABABA (all h)	5	24 × 29	2 (blue)	right, left, right, left: the simplest chevron
6H	ABCACBA (hcchcc)	7	24 × 43	4	two-layer slants turning at each h layer
Fibonacci	ABACBCACBABCA (chcchchcchcch)	13	24 × 85	4	never-repeating: slants of one and two steps in the Fibonacci order

The Fibonacci word is the one of #69 (c → ch, h → c), its first 13 letters. The swatches are given in full in the appendix. None has been knitted yet: they are an invitation, and photographs of the knitted swatches will follow. The swatches will test whether the bias rows place the fabric as Theorem 2 assumes at this gauge, and how the colours read.

7. Prior art

We looked for the idea (a knitted fabric as the exact edge-on slice of a close-packed stacking, placed by bias shifts, coloured by the Rule-of-Halves pieces) and for its parts. Sources checked:

- **DICTO's reference library** (302 items, searched by meaning with six queries: knitting mathematics; crochet hyperbolic plane; textile representation of crystal structure; stacking sequence polytype visualisation zigzag projection; close packing polytypes Zhdanov notation; Delaunay tetrahedra octahedra slice of close packing). No knitting source in the library. Nearest items: Krishnamurthy, Akleman et al. 2020 and 2022, space-filling tiles from woven fabric (weaving, not knitting, not stackings); Hart, Hansen and Kuhs 2021, Markov stacking of disordered ice (stacking notation); Ganesh and Nasr 2023, randomly stacked solids; Fedorov 1885 on layers of a lattice.
- **Mathematical knitting and crochet:** belcastro and Yackel (eds.), *Making Mathematics with Needlework* (2008), *Crafting by Concepts* (2011) and *Figuring Fibers* (2018), from publisher and library contents listings (surfaces, tori, knots and links, helix striping, the Chinese remainder theorem and stitch patterns); belcastro, "Every topological surface can be knit: a proof", J. Math. Arts 3 (2009) 67–83; Taimiņa, *Crocheting Adventures with Hyperbolic Planes* and Bridges 2022 "What I learned in 25 years crocheting hyperbolic planes"; Bridges papers: Taalman and Yackel 2020, "Wallpaper patterns for lattice designs"; Markande and Matsumoto 2020, "Knotty knits are tangles in tori"; "Knitting on helicoid scaffolds" (2024); Baker and Goldstine, crystallographic bead crochet (Bridges 2017); Morton and Grishanov, "Doubly periodic textile patterns" (arXiv 0806.2824). None treats close-packed stackings.
- **Knitted crystal models:** Alexander Crum Brown's 1883 model of rock salt from knitting needles and wool balls (University of Edinburgh collections) and his knitted pieces on interpenetrating surfaces: three-dimensional models, not a fabric that is a slice; Clare Bryden's Particulart knitted atoms and molecules; the Ravelry "Crystallography Socks" (inspired by crystallisation, cable charts); Philip Ball, "Weaving is believing", Chemistry World.
- **Bias knitting and stackings:** the Wikipedia article on bias knitting (chevrons as bias that reverses at intervals); web searches pairing bias knitting, chevrons, stacking faults,

fcc/hcp, polytypes and Zhdanov notation with knitting, crochet or textiles. Stacking diagrams of polytypes (Ramsdell's zigzag plots, as cited in the mineralogy literature) are drawings, not textiles.

Verdict. We found nowhere a knitted fabric used as an exact slice of a close-packed stacking, nor bias shifts used to place its layers, nor its colouring by the pieces between the balls. Knitted models of crystals (Crum Brown) and mathematical knitting of surfaces and lattices exist; the slice, the bias rule and the colouring are new as far as these sources show.

8. Conclusion

One plane, perpendicular to a row of touching balls, turns every close-packed stacking into a flat pattern of two shapes, triangles and rhombi, whose halves carry the colours of their layers in exactly equal dark and light shares; the c and h letters of the stacking become the knitter's "keep going" and "turn back". Status: theorems proved and checked exactly; generator done; five swatches charted, none knitted yet.

In the app. Kaleidohedra, HexOct (kaleidohedra.dictospheres.com/?open=hexoct:fib), opens the Fibonacci stacking, whose edge-on slice is swatch E.

References

[#57] DICTO, The Rule of Halves: the Holes of Every Close-Packed Stacking, 2026, DICTO DOI: 10.5281/zenodo.23647898. [#69] DICTO, The Hexagonality Law and the Fibonacci Stacking, 2026, DICTO DOI: 10.5281/zenodo.23721038. [#75] DICTO, Two Tiles for Every Stacking, 2026, DICTO DOI: 10.5281/zenodo.23737639. DICTO, *HexOct: The Hexagonality Law* (book), 2026, DICTO DOI: 10.5281/zenodo.23731359. Prior-art sources as listed in Section 7.

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Appendix: the five swatches

How to knit every swatch

- Stockinette. Stitches 1 and the last are edge stitches, worked in the chart colour, not slipped, so the edges show the bias steps.
- Colour changes: intarsia with small bobbins, twisting at every change; carry a colour along the back only where its next block is 1–2 stitches away.
- Every bias row keeps the stitch count: one decrease and one increase, each one stitch in from its edge. In the colour runs the increased and decreased stitches are counted as they finish; work each in the colour of its run.
- Bobble (MB), in the colour of the stitch. RS rows: (k1, yo, k1) into the stitch, turn, p3, turn, k3, turn, p3, turn, sl1-k2tog-psso. WS rows: (p1, yo, p1) into the stitch, turn, k3, turn, p3, turn, k3, turn, p3tog, and push it to the RS; if it sits badly, purl the stitch plain and sew a bobble or French knot there afterwards.
- Abbreviations: DG dark gold, LG light gold, DB dark blue, LB light blue. Runs are given in working order (RS from the last stitch to stitch 1, WS from stitch 1 to the last).

A. 4H: ABCBA (hchch), 24 stitches × 29 rows

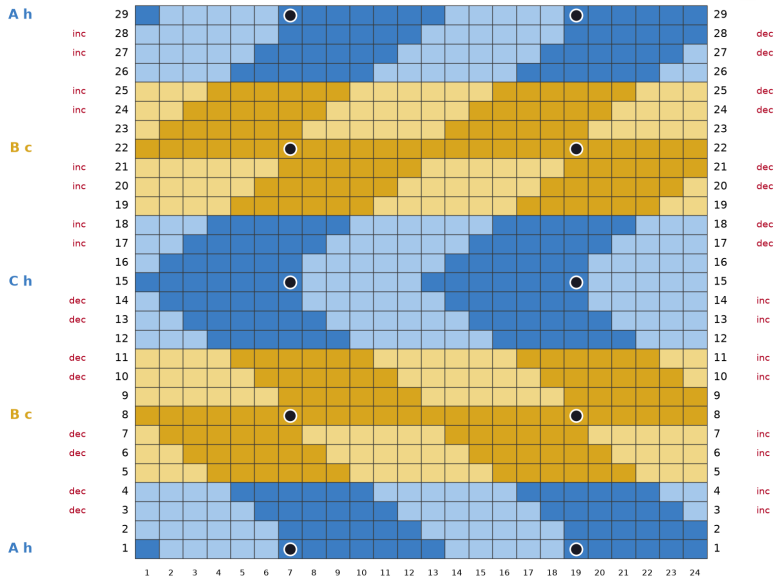
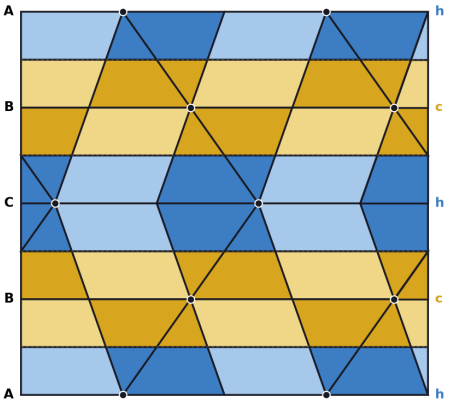


Chart. Seen from the RS, row 1 at the bottom; RS rows read right to left, WS rows left to right. Dots = bobbles. *dec*/*inc*: which edge of a bias row gets the decrease and which the increase.



The exact slice for this window (dots = ball centres; letters = layers; c gold, h blue; dark = tetrahedral, light = octahedral pieces).

DG dark gold: 192 sts · LG light gold: 144 sts · DB dark blue: 186 sts · LB light blue: 174 sts. Total 696 stitches.

Cast on 24 sts in DB; row 1 is a RS row. **Bias right** (decrease at the left edge, increase at the right, seen from the RS): rows 3, 4, 6, 7, 10, 11, 13, 14. **Bias left**: rows 17, 18, 20, 21, 24, 25, 27, 28. Net shift at the top: +0 stitches. **Bobbles** on RS rows 1, 15, 29; on WS rows 8, 22. Bind off in DB on the WS after row 29.

Row	Side	Colour runs (working order)	Bias (edge stitches)	Bobble
1	RS	6 DB, 5 LB, 7 DB, 5 LB, 1 DB		MB at st 7, 19
2	WS	6 LB, 6 DB, 6 LB, 6 DB		
3	RS	1 LB, 6 DB, 6 LB, 6 DB, 5 LB	**bias right**: k1, M1, … k2tog, k1	
4	WS	4 LB, 6 DB, 6 LB, 6 DB, 2 LB	**bias right**: p1, p2tog, … M1P, p1	
5	RS	3 LG, 6 DG, 6 LG, 6 DG, 3 LG		
6	WS	2 LG, 6 DG, 6 LG, 6 DG, 4 LG	**bias right**: p1, p2tog, … M1P, p1	
7	RS	5 LG, 6 DG, 6 LG, 6 DG, 1 LG	**bias right**: k1, M1, … k2tog, k1	
8	WS	24 DG		MB at st 7, 19

9	RS	6 DG, 6 LG, 6 DG, 6 LG		
10	WS	5 LG, 6 DG, 6 LG, 6 DG, 1 LG	**bias right**: p1, p2tog, … M1P, p1	
11	RS	2 LG, 6 DG, 6 LG, 6 DG, 4 LG	**bias right**: k1, M1, … k2tog, k1	
12	WS	3 LB, 6 DB, 6 LB, 6 DB, 3 LB		
13	RS	4 LB, 6 DB, 6 LB, 6 DB, 2 LB	**bias right**: k1, M1, … k2tog, k1	
14	WS	1 LB, 6 DB, 6 LB, 6 DB, 5 LB	**bias right**: p1, p2tog, … M1P, p1	
15	RS	5 LB, 7 DB, 5 LB, 7 DB		MB at st 7, 19
16	WS	1 LB, 6 DB, 6 LB, 6 DB, 5 LB		
17	RS	4 LB, 6 DB, 6 LB, 6 DB, 2 LB	**bias left**: k1, ssk, … M1, k1	
18	WS	3 LB, 6 DB, 6 LB, 6 DB, 3 LB	**bias left**: p1, M1P, … ssp, p1	
19	RS	2 LG, 6 DG, 6 LG, 6 DG, 4 LG		
20	WS	5 LG, 6 DG, 6 LG, 6 DG, 1 LG	**bias left**: p1, M1P, … ssp, p1	
21	RS	6 DG, 6 LG, 6 DG, 6 LG	**bias left**: k1, ssk, … M1, k1	
22	WS	24 DG		MB at st 7, 19
23	RS	5 LG, 6 DG, 6 LG, 6 DG, 1 LG		
24	WS	2 LG, 6 DG, 6 LG, 6 DG, 4 LG	**bias left**: p1, M1P, … ssp, p1	
25	RS	3 LG, 6 DG, 6 LG, 6 DG, 3 LG	**bias left**: k1, ssk, … M1, k1	
26	WS	4 LB, 6 DB, 6 LB, 6 DB, 2 LB		
27	RS	1 LB, 6 DB, 6 LB, 6 DB, 5 LB	**bias left**: k1, ssk, … M1, k1	
28	WS	6 LB, 6 DB, 6 LB, 6 DB	**bias left**: p1, M1P, … ssp, p1	
29	RS	6 DB, 5 LB, 7 DB, 5 LB, 1 DB		MB at st 7, 19

B. 3C: ABCABCA (ccccccc), 24 stitches × 43 rows

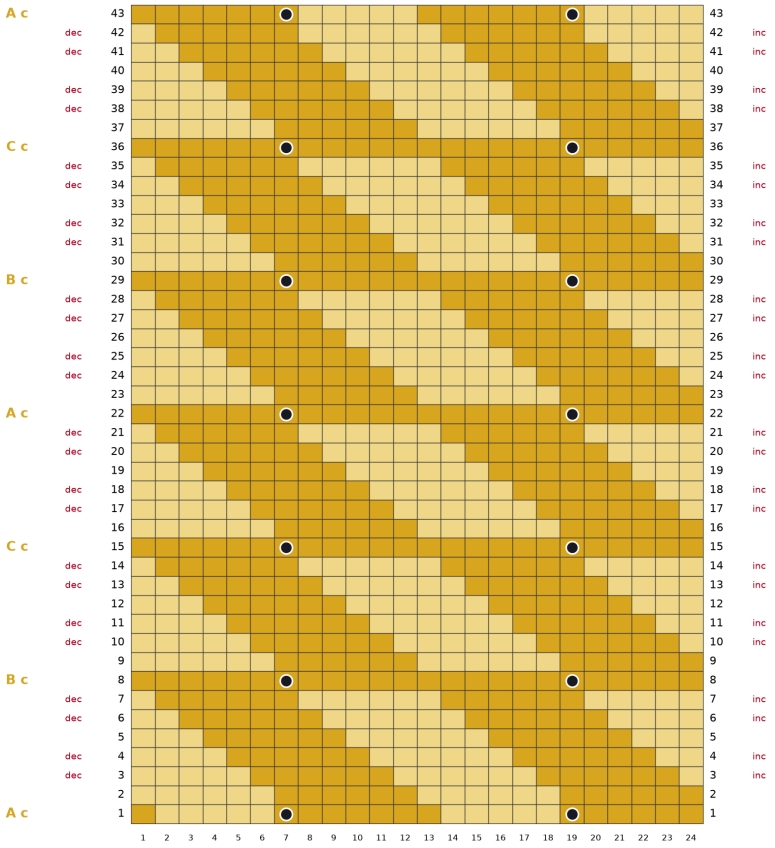
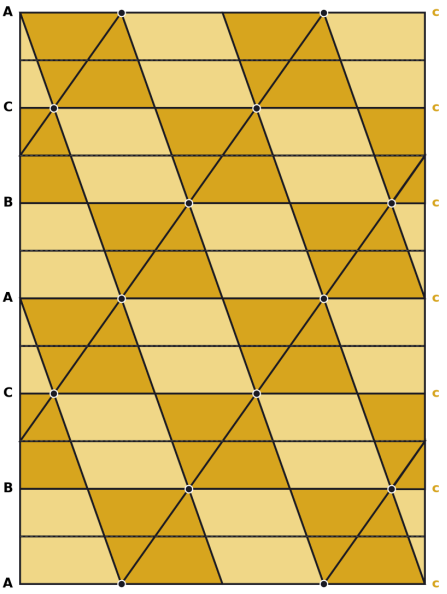


Chart. Seen from the RS, row 1 at the bottom; RS rows read right to left, WS rows left to right. Dots = bobbles. *dec*/*inc*: which edge of a bias row gets the decrease and which the increase.



The exact slice for this window (dots = ball centres; letters = layers; c gold, h blue; dark = tetrahedral, light = octahedral pieces).

DG dark gold: 580 sts · LG light gold: 452 sts. Total 1032 stitches.

Cast on 24 sts in DG; row 1 is a RS row. **Bias right** (decrease at the left edge, increase at the right, seen from the RS): rows 3, 4, 6, 7, 10, 11, 13, 14, 17, 18, 20, 21, 24, 25, 27, 28, 31, 32, 34, 35, 38, 39, 41, 42. **Bias left**: rows none. Net shift at the top: +24 stitches. **Bobbles** on RS rows 1, 15, 29, 43; on WS rows 8, 22, 36. Bind off in LG on the WS after row 43.

Row	Side	Colour runs (working order)	Bias (edge stitches)	Bobble
1	RS	6 DG, 5 LG, 7 DG, 5 LG, 1 DG		MB at st 7, 19
2	WS	6 LG, 6 DG, 6 LG, 6 DG		
3	RS	1 LG, 6 DG, 6 LG, 6 DG, 5 LG	**bias right**: k1, M1, … k2tog, k1	
4	WS	4 LG, 6 DG, 6 LG, 6 DG, 2 LG	**bias right**: p1, p2tog, … M1P, p1	
5	RS	3 LG, 6 DG, 6 LG, 6 DG, 3 LG		
6	WS	2 LG, 6 DG, 6 LG, 6 DG, 4 LG	**bias right**: p1, p2tog, … M1P, p1	
7	RS	5 LG, 6 DG, 6 LG, 6 DG, 1 LG	**bias right**: k1, M1, … k2tog, k1	
8	WS	24 DG		MB at st 7, 19
9	RS	6 DG, 6 LG, 6 DG, 6 LG		
10	WS	5 LG, 6 DG, 6 LG, 6 DG, 1 LG	**bias right**: p1, p2tog, … M1P, p1	
11	RS	2 LG, 6 DG, 6 LG, 6 DG, 4 LG	**bias right**: k1, M1, … k2tog, k1	
12	WS	3 LG, 6 DG, 6 LG, 6 DG, 3 LG		
13	RS	4 LG, 6 DG, 6 LG, 6 DG, 2 LG	**bias right**: k1, M1, … k2tog, k1	
14	WS	1 LG, 6 DG, 6 LG, 6 DG, 5 LG	**bias right**: p1, p2tog, … M1P, p1	
15	RS	24 DG		MB at st 7, 19
16	WS	6 LG, 6 DG, 6 LG, 6 DG		
17	RS	1 LG, 6 DG, 6 LG, 6 DG, 5 LG	**bias right**: k1, M1, … k2tog, k1	
18	WS	4 LG, 6 DG, 6 LG, 6 DG, 2 LG	**bias right**: p1, p2tog, … M1P, p1	
19	RS	3 LG, 6 DG, 6 LG, 6 DG, 3 LG		
20	WS	2 LG, 6 DG, 6 LG, 6 DG, 4 LG	**bias right**: p1, p2tog, … M1P, p1	

21	RS	5 LG, 6 DG, 6 LG, 6 DG, 1 LG	**bias right**: k1, M1, … k2tog, k1	
22	WS	24 DG		MB at st 7, 19
23	RS	6 DG, 6 LG, 6 DG, 6 LG		
24	WS	5 LG, 6 DG, 6 LG, 6 DG, 1 LG	**bias right**: p1, p2tog, … M1P, p1	
25	RS	2 LG, 6 DG, 6 LG, 6 DG, 4 LG	**bias right**: k1, M1, … k2tog, k1	
26	WS	3 LG, 6 DG, 6 LG, 6 DG, 3 LG		
27	RS	4 LG, 6 DG, 6 LG, 6 DG, 2 LG	**bias right**: k1, M1, … k2tog, k1	
28	WS	1 LG, 6 DG, 6 LG, 6 DG, 5 LG	**bias right**: p1, p2tog, … M1P, p1	
29	RS	24 DG		MB at st 7, 19
30	WS	6 LG, 6 DG, 6 LG, 6 DG		
31	RS	1 LG, 6 DG, 6 LG, 6 DG, 5 LG	**bias right**: k1, M1, … k2tog, k1	
32	WS	4 LG, 6 DG, 6 LG, 6 DG, 2 LG	**bias right**: p1, p2tog, … M1P, p1	
33	RS	3 LG, 6 DG, 6 LG, 6 DG, 3 LG		
34	WS	2 LG, 6 DG, 6 LG, 6 DG, 4 LG	**bias right**: p1, p2tog, … M1P, p1	
35	RS	5 LG, 6 DG, 6 LG, 6 DG, 1 LG	**bias right**: k1, M1, … k2tog, k1	
36	WS	24 DG		MB at st 7, 19
37	RS	6 DG, 6 LG, 6 DG, 6 LG		
38	WS	5 LG, 6 DG, 6 LG, 6 DG, 1 LG	**bias right**: p1, p2tog, … M1P, p1	
39	RS	2 LG, 6 DG, 6 LG, 6 DG, 4 LG	**bias right**: k1, M1, … k2tog, k1	
40	WS	3 LG, 6 DG, 6 LG, 6 DG, 3 LG		
41	RS	4 LG, 6 DG, 6 LG, 6 DG, 2 LG	**bias right**: k1, M1, … k2tog, k1	
42	WS	1 LG, 6 DG, 6 LG, 6 DG, 5 LG	**bias right**: p1, p2tog, … M1P, p1	
43	RS	5 LG, 7 DG, 5 LG, 7 DG		MB at st 7, 19

C. 2H: ABABA (hhhhh), 24 stitches × 29 rows

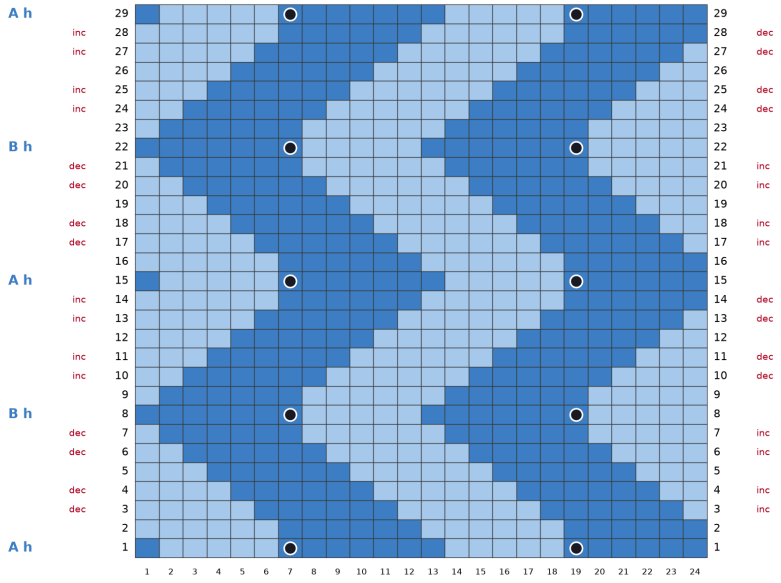
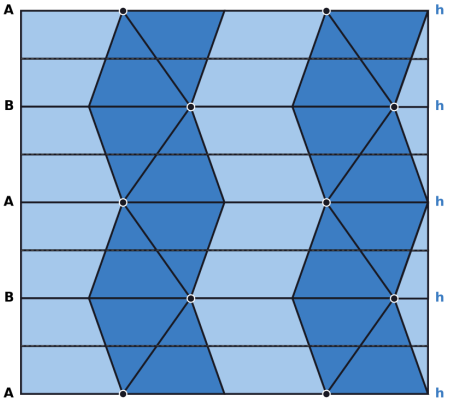


Chart. Seen from the RS, row 1 at the bottom; RS rows read right to left, WS rows left to right. Dots = bobbles. *dec*/*inc*: which edge of a bias row gets the decrease and which the increase.



The exact slice for this window (dots = ball centres; letters = layers; c gold, h blue; dark = tetrahedral, light = octahedral pieces).

DB dark blue: 358 sts · LB light blue: 338 sts. Total 696 stitches.

Cast on 24 sts in DB; row 1 is a RS row. **Bias right** (decrease at the left edge, increase at the right, seen from the RS): rows 3, 4, 6, 7, 17, 18, 20, 21. **Bias left**: rows 10, 11, 13, 14, 24, 25, 27, 28. Net shift at the top: +0 stitches. **Bobbles** on RS rows 1, 15, 29; on WS rows 8, 22. Bind off in DB on the WS after row 29.

Row	Side	Colour runs (working order)	Bias (edge stitches)	Bobble
1	RS	6 DB, 5 LB, 7 DB, 5 LB, 1 DB		MB at st 7, 19
2	WS	6 LB, 6 DB, 6 LB, 6 DB		
3	RS	1 LB, 6 DB, 6 LB, 6 DB, 5 LB	**bias right**: k1, M1, … k2tog, k1	
4	WS	4 LB, 6 DB, 6 LB, 6 DB, 2 LB	**bias right**: p1, p2tog, … M1P, p1	
5	RS	3 LB, 6 DB, 6 LB, 6 DB, 3 LB		
6	WS	2 LB, 6 DB, 6 LB, 6 DB, 4 LB	**bias right**: p1, p2tog, … M1P, p1	
7	RS	5 LB, 6 DB, 6 LB, 6 DB, 1 LB	**bias right**: k1, M1, … k2tog, k1	
8	WS	7 DB, 5 LB, 7 DB, 5 LB		MB at st 7, 19
9	RS	5 LB, 6 DB, 6 LB, 6 DB, 1 LB		
10	WS	2 LB, 6 DB, 6 LB, 6 DB, 4 LB	**bias left**: p1, M1P, … ssp, p1	
11	RS	3 LB, 6 DB, 6 LB, 6 DB, 3 LB	**bias left**: k1, ssk, … M1, k1	
12	WS	4 LB, 6 DB, 6 LB, 6 DB, 2 LB		
13	RS	1 LB, 6 DB, 6 LB, 6 DB, 5 LB	**bias left**: k1, ssk, … M1, k1	
14	WS	6 LB, 6 DB, 6 LB, 6 DB	**bias left**: p1, M1P, … ssp, p1	
15	RS	6 DB, 5 LB, 7 DB, 5 LB, 1 DB		MB at st 7, 19
16	WS	6 LB, 6 DB, 6 LB, 6 DB		
17	RS	1 LB, 6 DB, 6 LB, 6 DB, 5 LB	**bias right**: k1, M1, … k2tog, k1	
18	WS	4 LB, 6 DB, 6 LB, 6 DB, 2 LB	**bias right**: p1, p2tog, … M1P, p1	
19	RS	3 LB, 6 DB, 6 LB, 6 DB, 3 LB		
20	WS	2 LB, 6 DB, 6 LB, 6 DB, 4 LB	**bias right**: p1, p2tog, … M1P, p1	
21	RS	5 LB, 6 DB, 6 LB, 6 DB, 1 LB	**bias right**: k1, M1, … k2tog, k1	
22	WS	7 DB, 5 LB, 7 DB, 5 LB		MB at st 7, 19
23	RS	5 LB, 6 DB, 6 LB, 6 DB, 1 LB		
24	WS	2 LB, 6 DB, 6 LB, 6 DB, 4 LB	**bias left**: p1, M1P, … ssp, p1	
25	RS	3 LB, 6 DB, 6 LB, 6 DB, 3 LB	**bias left**: k1, ssk, … M1, k1	
26	WS	4 LB, 6 DB, 6 LB, 6 DB, 2 LB		
27	RS	1 LB, 6 DB, 6 LB, 6 DB, 5 LB	**bias left**: k1, ssk, … M1, k1	
28	WS	6 LB, 6 DB, 6 LB, 6 DB	**bias left**: p1, M1P, … ssp, p1	
29	RS	6 DB, 5 LB, 7 DB, 5 LB, 1 DB		MB at st 7, 19

D. 6H: ABCACBA (hcchcch), 24 stitches × 43 rows

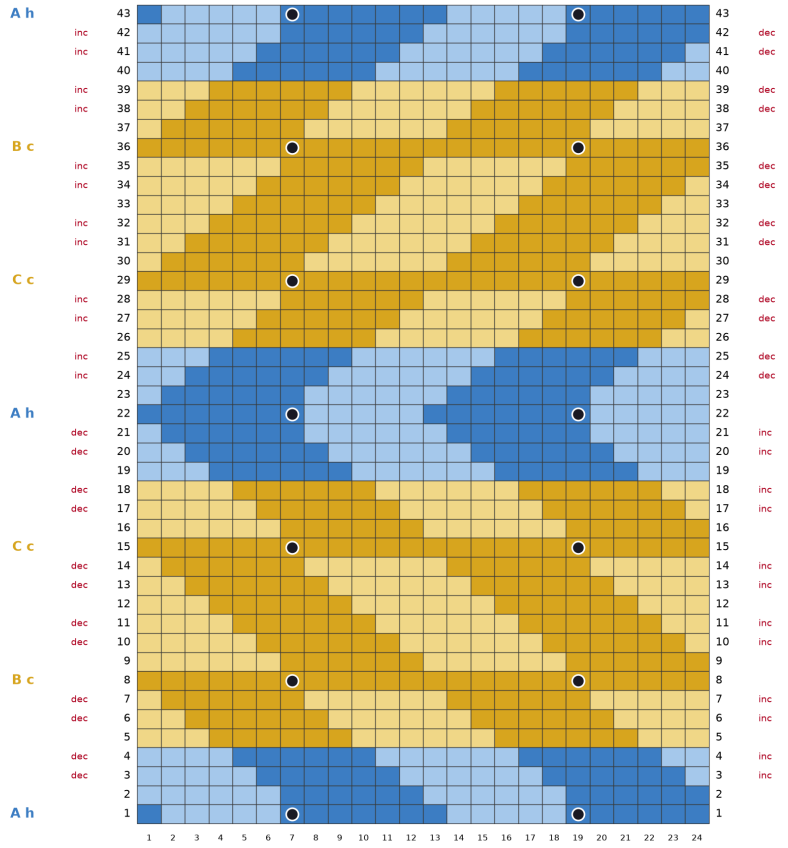
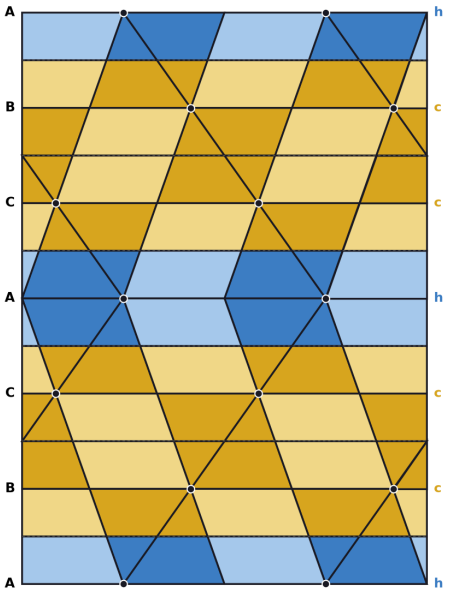


Chart. Seen from the RS, row 1 at the bottom; RS rows read right to left, WS rows left to right. Dots = bobbles. *dec*/*inc*: which edge of a bias row gets the decrease and which the increase.



The exact slice for this window (dots = ball centres; letters = layers; c gold, h blue; dark = tetrahedral, light = octahedral pieces).

DG dark gold: 384 sts · LG light gold: 288 sts · DB dark blue: 186 sts · LB light blue: 174 sts. Total 1032 stitches.

Cast on 24 sts in DB; row 1 is a RS row. **Bias right** (decrease at the left edge, increase at the right, seen from the RS): rows 3, 4, 6, 7, 10, 11, 13, 14, 17, 18, 20, 21. **Bias left**: rows 24, 25, 27, 28, 31, 32, 34, 35, 38, 39, 41, 42. Net shift at the top: +0 stitches. **Bobbles** on RS rows 1, 15, 29, 43; on WS rows 8, 22, 36. Bind off in DB on the WS after row 43.

Row	Side	Colour runs (working order)	Bias (edge stitches)	Bobble
1	RS	6 DB, 5 LB, 7 DB, 5 LB, 1 DB		MB at st 7, 19
2	WS	6 LB, 6 DB, 6 LB, 6 DB		
3	RS	1 LB, 6 DB, 6 LB, 6 DB, 5 LB	**bias right**: k1, M1, … k2tog, k1	
4	WS	4 LB, 6 DB, 6 LB, 6 DB, 2 LB	**bias right**: p1, p2tog, … M1P, p1	
5	RS	3 LG, 6 DG, 6 LG, 6 DG, 3 LG		
6	WS	2 LG, 6 DG, 6 LG, 6 DG, 4 LG	**bias right**: p1, p2tog, … M1P, p1	
7	RS	5 LG, 6 DG, 6 LG, 6 DG, 1 LG	**bias right**: k1, M1, … k2tog, k1	
8	WS	24 DG		MB at st 7, 19
9	RS	6 DG, 6 LG, 6 DG, 6 LG		
10	WS	5 LG, 6 DG, 6 LG, 6 DG, 1 LG	**bias right**: p1, p2tog, … M1P, p1	
11	RS	2 LG, 6 DG, 6 LG, 6 DG, 4 LG	**bias right**: k1, M1, … k2tog, k1	
12	WS	3 LG, 6 DG, 6 LG, 6 DG, 3 LG		
13	RS	4 LG, 6 DG, 6 LG, 6 DG, 2 LG	**bias right**: k1, M1, … k2tog, k1	
14	WS	1 LG, 6 DG, 6 LG, 6 DG, 5 LG	**bias right**: p1, p2tog, … M1P, p1	
15	RS	24 DG		MB at st 7, 19
16	WS	6 LG, 6 DG, 6 LG, 6 DG		
17	RS	1 LG, 6 DG, 6 LG, 6 DG, 5 LG	**bias right**: k1, M1, … k2tog, k1	
18	WS	4 LG, 6 DG, 6 LG, 6 DG, 2 LG	**bias right**: p1, p2tog, … M1P, p1	
19	RS	3 LB, 6 DB, 6 LB, 6 DB, 3 LB		
20	WS	2 LB, 6 DB, 6 LB, 6 DB, 4 LB	**bias right**: p1, p2tog, … M1P, p1	

21	RS	5 LB, 6 DB, 6 LB, 6 DB, 1 LB	**bias right**: k1, M1, … k2tog, k1	
22	WS	7 DB, 5 LB, 7 DB, 5 LB		MB at st 7, 19
23	RS	5 LB, 6 DB, 6 LB, 6 DB, 1 LB		
24	WS	2 LB, 6 DB, 6 LB, 6 DB, 4 LB	**bias left**: p1, M1P, … ssp, p1	
25	RS	3 LB, 6 DB, 6 LB, 6 DB, 3 LB	**bias left**: k1, ssk, … M1, k1	
26	WS	4 LG, 6 DG, 6 LG, 6 DG, 2 LG		
27	RS	1 LG, 6 DG, 6 LG, 6 DG, 5 LG	**bias left**: k1, ssk, … M1, k1	
28	WS	6 LG, 6 DG, 6 LG, 6 DG	**bias left**: p1, M1P, … ssp, p1	
29	RS	24 DG		MB at st 7, 19
30	WS	1 LG, 6 DG, 6 LG, 6 DG, 5 LG		
31	RS	4 LG, 6 DG, 6 LG, 6 DG, 2 LG	**bias left**: k1, ssk, … M1, k1	
32	WS	3 LG, 6 DG, 6 LG, 6 DG, 3 LG	**bias left**: p1, M1P, … ssp, p1	
33	RS	2 LG, 6 DG, 6 LG, 6 DG, 4 LG		
34	WS	5 LG, 6 DG, 6 LG, 6 DG, 1 LG	**bias left**: p1, M1P, … ssp, p1	
35	RS	6 DG, 6 LG, 6 DG, 6 LG	**bias left**: k1, ssk, … M1, k1	
36	WS	24 DG		MB at st 7, 19
37	RS	5 LG, 6 DG, 6 LG, 6 DG, 1 LG		
38	WS	2 LG, 6 DG, 6 LG, 6 DG, 4 LG	**bias left**: p1, M1P, … ssp, p1	
39	RS	3 LG, 6 DG, 6 LG, 6 DG, 3 LG	**bias left**: k1, ssk, … M1, k1	
40	WS	4 LB, 6 DB, 6 LB, 6 DB, 2 LB		
41	RS	1 LB, 6 DB, 6 LB, 6 DB, 5 LB	**bias left**: k1, ssk, … M1, k1	
42	WS	6 LB, 6 DB, 6 LB, 6 DB	**bias left**: p1, M1P, … ssp, p1	
43	RS	6 DB, 5 LB, 7 DB, 5 LB, 1 DB		MB at st 7, 19

E. Fibonacci: ABACBCACBABCA (chcchchcchcch), 24 stitches × 85 rows

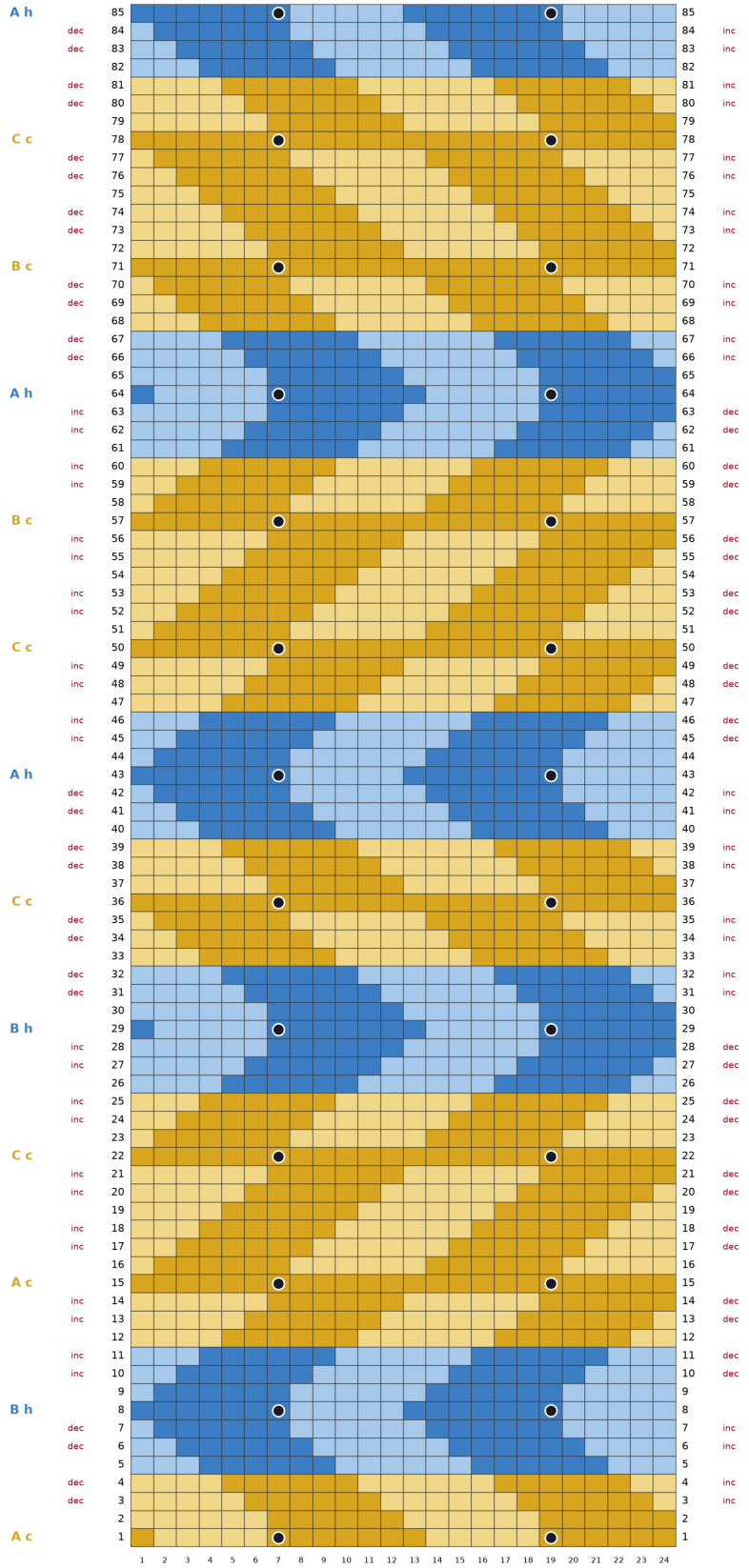
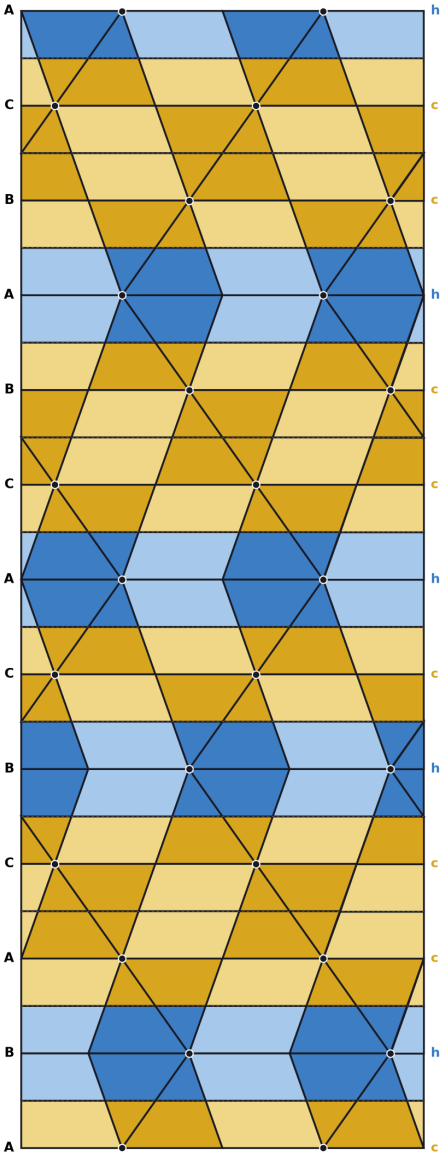


Chart. Seen from the RS, row 1 at the bottom; RS rows read right to left, WS rows left to right. Dots = bobbles. *dec*/*inc*: which edge of a bias row gets the decrease and which the increase.



The exact slice for this window (dots = ball centres; letters = layers; c gold, h blue; dark = tetrahedral, light = octahedral pieces).

DG dark gold: 722 sts · LG light gold: 550 sts · DB dark blue: 394 sts · LB light blue: 374 sts. Total 2040 stitches.

Cast on 24 sts in DG; row 1 is a RS row. **Bias right** (decrease at the left edge, increase at the right, seen from the RS): rows 3, 4, 6, 7, 31, 32, 34, 35, 38, 39, 41, 42, 66, 67, 69, 70, 73, 74, 76, 77, 80, 81, 83, 84. **Bias left**: rows 10, 11, 13, 14, 17, 18, 20, 21, 24, 25, 27, 28, 45, 46, 48, 49, 52, 53, 55, 56, 59, 60, 62, 63. Net shift at the top: +0 stitches. **Bobbles** on RS rows 1, 15, 29, 43, 57, 71, 85; on WS rows 8, 22, 36, 50, 64, 78. Bind off in LB on the WS after row 85.

Row	Side	Colour runs (working order)	Bias (edge stitches)	Bobble
1	RS	6 DG, 5 LG, 7 DG, 5 LG, 1 DG		MB at st 7, 19
2	WS	6 LG, 6 DG, 6 LG, 6 DG		
3	RS	1 LG, 6 DG, 6 LG, 6 DG, 5 LG	**bias right**: k1, M1, … k2tog, k1	
4	WS	4 LG, 6 DG, 6 LG, 6 DG, 2 LG	**bias right**: p1, p2tog, … M1P, p1	
5	RS	3 LB, 6 DB, 6 LB, 6 DB, 3 LB		
6	WS	2 LB, 6 DB, 6 LB, 6 DB, 4 LB	**bias right**: p1, p2tog, … M1P, p1	
7	RS	5 LB, 6 DB, 6 LB, 6 DB, 1 LB	**bias right**: k1, M1, … k2tog, k1	
8	WS	7 DB, 5 LB, 7 DB, 5 LB		MB at st 7, 19
9	RS	5 LB, 6 DB, 6 LB, 6 DB, 1 LB		
10	WS	2 LB, 6 DB, 6 LB, 6 DB, 4 LB	**bias left**: p1, M1P, … ssp, p1	
11	RS	3 LB, 6 DB, 6 LB, 6 DB, 3 LB	**bias left**: k1, ssk, … M1, k1	
12	WS	4 LG, 6 DG, 6 LG, 6 DG, 2 LG		
13	RS	1 LG, 6 DG, 6 LG, 6 DG, 5 LG	**bias left**: k1, ssk, … M1, k1	
14	WS	6 LG, 6 DG, 6 LG, 6 DG	**bias left**: p1, M1P, … ssp, p1	
15	RS	24 DG		MB at st 7, 19
16	WS	1 LG, 6 DG, 6 LG, 6 DG, 5 LG		
17	RS	4 LG, 6 DG, 6 LG, 6 DG, 2 LG	**bias left**: k1, ssk, … M1, k1	
18	WS	3 LG, 6 DG, 6 LG, 6 DG, 3 LG	**bias left**: p1, M1P, … ssp, p1	
19	RS	2 LG, 6 DG, 6 LG, 6 DG, 4 LG		
20	WS	5 LG, 6 DG, 6 LG, 6 DG, 1 LG	**bias left**: p1, M1P, … ssp, p1	
21	RS	6 DG, 6 LG, 6 DG, 6 LG	**bias left**: k1, ssk, … M1, k1	
22	WS	24 DG		MB at st 7, 19
23	RS	5 LG, 6 DG, 6 LG, 6 DG, 1 LG		
24	WS	2 LG, 6 DG, 6 LG, 6 DG, 4 LG	**bias left**: p1, M1P, … ssp, p1	
25	RS	3 LG, 6 DG, 6 LG, 6 DG, 3 LG	**bias left**: k1, ssk, … M1, k1	
26	WS	4 LB, 6 DB, 6 LB, 6 DB, 2 LB		
27	RS	1 LB, 6 DB, 6 LB, 6 DB, 5 LB	**bias left**: k1, ssk, … M1, k1	
28	WS	6 LB, 6 DB, 6 LB, 6 DB	**bias left**: p1, M1P, … ssp, p1	
29	RS	6 DB, 5 LB, 7 DB, 5 LB, 1 DB		MB at st 7, 19
30	WS	6 LB, 6 DB, 6 LB, 6 DB		
31	RS	1 LB, 6 DB, 6 LB, 6 DB, 5 LB	**bias right**: k1, M1, … k2tog, k1	
32	WS	4 LB, 6 DB, 6 LB, 6 DB, 2 LB	**bias right**: p1, p2tog, … M1P, p1	
33	RS	3 LG, 6 DG, 6 LG, 6 DG, 3 LG		
34	WS	2 LG, 6 DG, 6 LG, 6 DG, 4 LG	**bias right**: p1, p2tog, … M1P, p1	
35	RS	5 LG, 6 DG, 6 LG, 6 DG, 1 LG	**bias right**: k1, M1, … k2tog, k1	
36	WS	24 DG		MB at st 7, 19
37	RS	6 DG, 6 LG, 6 DG, 6 LG		
38	WS	5 LG, 6 DG, 6 LG, 6 DG, 1 LG	**bias right**: p1, p2tog, … M1P, p1	
39	RS	2 LG, 6 DG, 6 LG, 6 DG, 4 LG	**bias right**: k1, M1, … k2tog, k1	
40	WS	3 LB, 6 DB, 6 LB, 6 DB, 3 LB		
41	RS	4 LB, 6 DB, 6 LB, 6 DB, 2 LB	**bias right**: k1, M1, … k2tog, k1	
42	WS	1 LB, 6 DB, 6 LB, 6 DB, 5 LB	**bias right**: p1, p2tog, … M1P, p1	
43	RS	5 LB, 7 DB, 5 LB, 7 DB		MB at st 7, 19
44	WS	1 LB, 6 DB, 6 LB, 6 DB, 5 LB		
45	RS	4 LB, 6 DB, 6 LB, 6 DB, 2 LB	**bias left**: k1, ssk, … M1, k1	
46	WS	3 LB, 6 DB, 6 LB, 6 DB, 3 LB	**bias left**: p1, M1P, … ssp, p1	
47	RS	2 LG, 6 DG, 6 LG, 6 DG, 4 LG		
48	WS	5 LG, 6 DG, 6 LG, 6 DG, 1 LG	**bias left**: p1, M1P, … ssp, p1	
49	RS	6 DG, 6 LG, 6 DG, 6 LG	**bias left**: k1, ssk, … M1, k1	
50	WS	24 DG		MB at st 7, 19
51	RS	5 LG, 6 DG, 6 LG, 6 DG, 1 LG		

52	WS	2 LG, 6 DG, 6 LG, 6 DG, 4 LG	**bias left**: p1, M1P, … ssp, p1	
53	RS	3 LG, 6 DG, 6 LG, 6 DG, 3 LG	**bias left**: k1, ssk, … M1, k1	
54	WS	4 LG, 6 DG, 6 LG, 6 DG, 2 LG		
55	RS	1 LG, 6 DG, 6 LG, 6 DG, 5 LG	**bias left**: k1, ssk, … M1, k1	
56	WS	6 LG, 6 DG, 6 LG, 6 DG	**bias left**: p1, M1P, … ssp, p1	
57	RS	24 DG		MB at st 7, 19
58	WS	1 LG, 6 DG, 6 LG, 6 DG, 5 LG		
59	RS	4 LG, 6 DG, 6 LG, 6 DG, 2 LG	**bias left**: k1, ssk, … M1, k1	
60	WS	3 LG, 6 DG, 6 LG, 6 DG, 3 LG	**bias left**: p1, M1P, … ssp, p1	
61	RS	2 LB, 6 DB, 6 LB, 6 DB, 4 LB		
62	WS	5 LB, 6 DB, 6 LB, 6 DB, 1 LB	**bias left**: p1, M1P, … ssp, p1	
63	RS	6 DB, 6 LB, 6 DB, 6 LB	**bias left**: k1, ssk, … M1, k1	
64	WS	1 DB, 5 LB, 7 DB, 5 LB, 6 DB		MB at st 7, 19
65	RS	6 DB, 6 LB, 6 DB, 6 LB		
66	WS	5 LB, 6 DB, 6 LB, 6 DB, 1 LB	**bias right**: p1, p2tog, … M1P, p1	
67	RS	2 LB, 6 DB, 6 LB, 6 DB, 4 LB	**bias right**: k1, M1, … k2tog, k1	
68	WS	3 LG, 6 DG, 6 LG, 6 DG, 3 LG		
69	RS	4 LG, 6 DG, 6 LG, 6 DG, 2 LG	**bias right**: k1, M1, … k2tog, k1	
70	WS	1 LG, 6 DG, 6 LG, 6 DG, 5 LG	**bias right**: p1, p2tog, … M1P, p1	
71	RS	24 DG		MB at st 7, 19
72	WS	6 LG, 6 DG, 6 LG, 6 DG		
73	RS	1 LG, 6 DG, 6 LG, 6 DG, 5 LG	**bias right**: k1, M1, … k2tog, k1	
74	WS	4 LG, 6 DG, 6 LG, 6 DG, 2 LG	**bias right**: p1, p2tog, … M1P, p1	
75	RS	3 LG, 6 DG, 6 LG, 6 DG, 3 LG		
76	WS	2 LG, 6 DG, 6 LG, 6 DG, 4 LG	**bias right**: p1, p2tog, … M1P, p1	
77	RS	5 LG, 6 DG, 6 LG, 6 DG, 1 LG	**bias right**: k1, M1, … k2tog, k1	
78	WS	24 DG		MB at st 7, 19
79	RS	6 DG, 6 LG, 6 DG, 6 LG		
80	WS	5 LG, 6 DG, 6 LG, 6 DG, 1 LG	**bias right**: p1, p2tog, … M1P, p1	
81	RS	2 LG, 6 DG, 6 LG, 6 DG, 4 LG	**bias right**: k1, M1, … k2tog, k1	
82	WS	3 LB, 6 DB, 6 LB, 6 DB, 3 LB		
83	RS	4 LB, 6 DB, 6 LB, 6 DB, 2 LB	**bias right**: k1, M1, … k2tog, k1	
84	WS	1 LB, 6 DB, 6 LB, 6 DB, 5 LB	**bias right**: p1, p2tog, … M1P, p1	
85	RS	5 LB, 7 DB, 5 LB, 7 DB		MB at st 7, 19